

SMART BANKING · TECHNICAL TERMS OF REFERENCE

Credit scoring explained

and limit engines

Time: 4–5 hours

Score: 100 (50 code + 50 live)

1 Problem

The decision to grant a loan or limit to a client is made in 30 seconds, but both the regulator and the client require an answer to the question "why was it denied?" **with the reason for the decision.**

Your task: Build a system that issues a loan decision based on an application, explaining each decision with factors.

2 Information provided to you

Folder: `datasetlar/a2_credit/`



Download dataset (.zip)

717.3 KB · with answer key · for self-testing

This is **an open sample dataset** . On day 2, judges will check on another (closed) dataset — in the same format, with different values.

File	What's up?
<code>applicants.csv</code>	Applicants: age, employment, length of service, declared income.
<code>monthly_flows.csv</code>	12 months of income/expense/cash withdrawals.
<code>existing_loans.csv</code>	Existing loans and delinquency history.
<code>applications.csv</code>	Applications. The result is 80% filled (train), 20% empty (test).
<code>_javob_kaliti/test_natijalari.csv</code>	Only you have the real results of the test applications and PD.

3 What to build (MVP framework)

MANDATORY

Application form (client side) + underwriter panel (bank side).

Scorecard calculation: based on transaction history, revenue stability, and current load.

The contribution of factors to each decision is indicated (why +18 points, why -25 points).

BONUS (IF TIME PERMITS)

"What-if" simulator: what will be the limit if I increase my income?

Scorecard version comparison screen.

MVP LIMIT

1 application form + 1 underwriter screen is enough. Give time to proper calculation rather than pretty graphics.

4 Mandatory algorithms (2 take — have 2.5 balls)

DTI/PTI account

Calculate the debt-to-income ratio (current monthly payments / monthly income).

Cash-flow tahlili

Calculate the median and variability (CV or std) of monthly income — stable income is less risky.

BONUS ALGORITHMS (+1 POINT EACH)

Scorecard / binning

Divide the characters into buckets with WOE (Weight of Evidence) or IV (Information Value).

Model

Logistic regression or decision tree - coefficients must be interpreted.

Limit search

Binary search bilan maksimal limitni toping.

TOPIC TRAP — BE SURE TO READ THIS

TRAP

The reason for the decision must be stated. If the rejection is without a reason, the main requirement of the topic is not met.

The contribution of factors should be calculated: for example, "–25 points for high DTI, +10 points for long work experience." Showing only the final score and hiding the "why" is like falling into a trap.

5 Base requirement

Application → decision → history chain; immutable decision log.

Scorecard versioned (SCD Type 2 or valid_from/valid_to) — so that old decisions can be recalculated with the old version.

RESULT FILE FORMAT — IMPORTANT FOR AUTOMATIC CHECKING

Fayl: `natija/kredit_qarorlari.csv` (nomida "qaror", "score" yoki "decision" so'zi bo'lishi shart)

Ustun	Tavsif	Shart
<code>application_id</code>	Test arizasining ID si (barcha test arizalar bo'lishi kerak)	✓
<code>score / pd / risk</code>	Sizning modelingiz chiqargan ball yoki defolt ehtimoli	✓
<code>sabab</code>	Asosiy omil (ixtiyoriy, bonus)	—

Faylga barcha test arizalarni yozing (faqat ba'zilarini emas) — baholovchi AUC hisoblash uchun to'liq ro'yxat kerak.

6 Topshirish checklisti

- ☐ README.md — ishga tushirish buyrug'i
- ☐ DECISIONS.md — 2 majburiy algoritm: nega shunday, Big-O, chekka holat, ishlatilgan pattern
- ☐ `natija/kredit_qarorlari.csv` — belgilangan formatda
- ☐ Kod ishlaydi: `docker-compose up` yoki `make run`
- ☐ Kamida bitta avtomatik test

7 Boshlash uchun birinchi 30 daqiqa

- 1 `_javob_kaliti/xulosa.json` ni o'qing — train/test taqsimotini tushunib oling.
- 2 Applicants + monthly_flows ni birlashtirib, DTI va CV ni hisoblang.
- 3 Eng sodda skoring formulani (masalan chiziqli kombinatsiya) darhol yozib, natija faylini chiqaring.

To'liq ballash tizimi va boshqa mavzular — **ballash-tizimi.md** va hakam tomonidan tarqatilgan boshqa hujjatlarda.
Savol bo'lsa — hakamga murojaat qiling.